Customer Churn Analysis Demo

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1 IBM Watson Customer Churn Data Set

The `wu_churn` data set is a collection of IBM Watson customer data collected in September 2018. The data contains a factor variable of whether or not the customer has churned, and other related information about the customer.

```
# Install and load the dataset
# install.packages("modeldata")
library(modeldata)
data(wa_churn)
str(wa_churn)
```

```
Classes 'tbl_df', 'tbl' and 'data.frame':   7043 obs. of  20 variables:
 $ churn            : Factor w/ 2 levels "Yes","No": 2 2 1 2 1 1 2 2 1 2 ...
 $ female           : num  1 0 0 0 1 1 0 1 1 0 ...
 $ senior_citizen   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ partner          : num  1 0 0 0 0 0 0 0 1 0 ...
 $ dependents       : num  0 0 0 0 0 0 1 0 0 1 ...
 $ tenure           : int  1 34 2 45 2 8 22 10 28 62 ...
 $ phone_service    : num  0 1 1 0 1 1 1 0 1 1 ...
```

```
$ multiple_lines   : Factor w/ 3 levels "No","No phone service",..: 2 1 1 2 1 3 3 2 3 1 ...
$ internet_service : Factor w/ 3 levels "DSL","Fiber optic",..: 1 1 1 1 2 2 2 1 2 1 ...
$ online_security  : Factor w/ 3 levels "No","No internet service",..: 1 3 3 3 1 1 1 3 1 3
$ online_backup    : Factor w/ 3 levels "No","No internet service",..: 3 1 3 1 1 1 3 1 1 3
$ device_protection: Factor w/ 3 levels "No","No internet service",..: 1 3 1 3 1 3 1 1 3 1
$ tech_support     : Factor w/ 3 levels "No","No internet service",..: 1 1 1 3 1 1 1 1 3 1
$ streaming_tv     : Factor w/ 3 levels "No","No internet service",..: 1 1 1 1 1 3 3 1 3 1
$ streaming_movies : Factor w/ 3 levels "No","No internet service",..: 1 1 1 1 1 3 1 1 3 1
$ contract         : Factor w/ 3 levels "Month-to-month",..: 1 2 1 2 1 1 1 1 1 2 ...
$ paperless_billing: num  1 0 1 0 1 1 1 0 1 0 ...
$ payment_method   : Factor w/ 4 levels "Bank transfer (automatic)",..: 3 4 4 1 3 3 2 4 3 1
$ monthly_charges  : num  29.9 57 53.9 42.3 70.7 ...
$ total_charges    : num  29.9 1889.5 108.2 1840.8 151.7 ...
```

2 Report Outcome

The goal of this report is to use the `wu_churn` data set to compare the performance between two models trained to predict whether a customer will churn based on their observed characteristics. Ultimately, this will show quantitative methods of scoring model performance and derive metrics for proving the strongest predictor choice.

3 Data Cleaning

Convert